ECE2800J

Programming and Introductory Data Structures

I/O Streams

Learning Objectives:

Understand I/O

Know how to read/write from standard input/output, files, and strings

I/O

- **I/O Streams**
  - **Overview**
  - **Output Stream cout**
  - Input Stream cin
  - File Stream
  - String Stream

Input/Output

Streams

- A popular model for how input and output is done in computer systems is centered around the notion of a **stream**.
- A stream is just a sequence of data with functions to put data into one end, and take them out of the other.

```
cin >> a;
```

Input/Output

Streams

- Typical streams:

```
keyboard   →   program
display    ←   program
file       →   program
file       ←   program
string     →   program
string     ←   program
```

- In C++, streams are **unidirectional**.
- Data is always passed through the stream in one direction.
- If you want to read and write data to the same file or device, you need two streams.

Input/Output

Streams

- In general, there are two kinds of stream data: **characters** and **binary data**.
- Characters are usually used for:
  - Communicating between your program and a keyboard or screen.
  - Reading and writing files.
- In addition to text, files can contain arbitrary **binary** data.
  - It is usually much more efficient than character representation.
  - However, it is hard to understand and debug.

- We'll talk about **character streams** here.

Outline

- I/O Streams
  - Overview
  - Output Stream cout
  - Input Stream cin
  - File Stream
  - String Stream

Output Stream: cout

```
cout << "Hello, world!\n";
```

- Output to screen.
- The << is called the **insertion operator**, and is used to insert things into the output stream.
  - It knows how to **convert** all of the other standard data types to **characters** before inserting them into the stream.

```
int foo = 42;
cout << foo << endl;
```

  - Can be cascaded

```
cout << foo << " " << bar << endl;
```

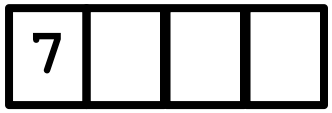
Print with Fixed Field Width

```
cout << foo << setw(4) << bar << endl;
```

- Here the `setw()` manipulator sets the width of the **following** number to the specified number of positions and **right-aligns** the number within that field.
- It pads with spaces.

right align

left align



- If you want to use `setw()` , you should

```
#include <iomanip>
```

Alternative Output Streams

- You can also use the Linux I/O **redirection** facility to move the output end of the stream from screen to a file:

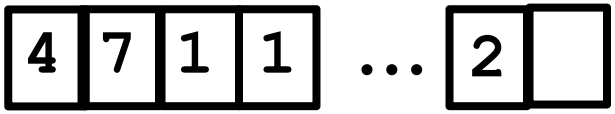
```
$ ./hello > foo
```

- This connects the output end of the `cout` stream to the file "`foo`".

- There is another output stream object defined by the `iostream` library called `cerr`.
- This stream is identical in most respects to the `cout` stream; in particular, its default output is also the screen.
- By convention, programs use the `cerr` stream for **error messages**.

Output: Buffering

- I/O in C++ is **buffered**.
- This means output inserted into an output stream is saved by the underlying operating system (in a region of memory called a **buffer**).



- The content in the buffer is written to the output only when specific actions are taken.

Output: Buffering

- The buffer content is written to the output only when:
  - A newline (e.g., `endl` or '`\n`') is inserted into the stream. E.g.,

    ```
    cout << "ok" << endl;
    ```

  - The buffer is explicitly flushed. E.g.,

    ```
    cout << "ok" << flush;
    ```

  - The buffer becomes full
  - The program decides to read from `cin`
  - The program exits
- Once the buffer content is written to the output, the buffer is **cleaned**
- If some content is not printed out, it may be still in the buffer
- In contrast, output sent to `cerr` is not buffered

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  - String Stream

Input Stream: cin

- `cin >> foo;`
  - Takes input from keyboard
  - >> is called the **extraction operator**, and is used to extract things from the input stream.
  - Knows how to convert the characters you type into values of simple types and strings.

Which Statements Are True?

- Given the following code, select all the correct answers.
  - Assume inputs is: `42 3.14 four score\n`

```
int foo;
double bar;
string baz;
cin >> foo >> bar >> baz;
```

- **A.** foo is "42"
- **B.** bar is "3"
- **C.** baz is "four"
- **D.** baz is "four score"

getline()

- In the previous example, `baz` is just "four"! How to get `baz` as "`four score`"?
- If you need to read strings including **blanks** or **tabs**, use the `getline()` function:

```
cin >> foo >> bar;
getline(cin, baz);
```

Assume inputs is:
`42 3.14 four score\n`

- `getline()` reads all characters **up to but not including** the next newline and puts them into the **string variable**, and then **discards the newline**
- But `baz` is "`  four score`"; it keeps the leading space

get()

- The `get()` function reads **a single** character, whitespace or newlines:

```
char ch;
cin.get(ch); // Extracts a character
//from cin stream and stores it in ch
```

- So, we can accomplish what we'd hoped to accomplish by:

```
cin >> foo >> bar;
cin.get(ch);
getline(cin, baz);
```

- This makes `baz` "`four score`".

Assume inputs is:
`42 3.14 four score\n`

The three methods have such different syntax.
However, the three methods can be freely intermixed.

Input: Buffering

- Like `cout`, `cin` is **buffered**.
- Characters typed (which are to be gathered by `cin`) are stored in a buffer **until the enter key** is pressed.
- The characters are then made available to the program as a group.
- This also allows for greater efficiency, and it lets you correct errors before your program sees them (i.e. you can go back and fix something you typed wrong).

Alternative Input Streams

- You can use the Linux I/O **redirection** facility to move the input end of the stream from the keyboard to a file:

```
$ prog < foo
```

- When doing this, remember that the input will not appear on your screen since you did not enter it on the keyboard.
  - This makes funny-looking output, as the input is not echoed.

Failed Input Streams

- The extraction operator will fail if inappropriate data is given to it.
- For example, if:

```
int foo;
cin >> foo;
```

  is presented with:

```
42abc\n
```

  the attempted conversion will succeed, up to the point of the "`a`", i.e., `foo = 42`

- The stream will be left with "`abc\n`" in it.

Failed Input Streams

```
int foo;
cin >> foo;
```

- However, if you present it with something that **does not** begin with a digit, like:

  `abc`

  then the stream will enter a **failed** state.

- You can test the state of a stream by using it where a bool is expected:
  - For example, `if(cin) {…}` `while(cin) {…}`
  - It returns **true** if it is **good**, false otherwise.

- A failed input stream will resist all attempts to extract more data from it

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File Streams

- Why use files?
  - Files allow you to store data permanently!
  - Data output to a file lasts after the program ends
  - An input file can be used over and over. No typing of data again and again for testing
- File stream: I/O between file and program

- Linux has I/O redirection facility. Then, why use file streams?
  - E.g., when you need to write to two files

Using File Streams

- `#include <fstream>`
- Declare an **input** file stream object

```
ifstream iFile;
```

- Declare an **output** file stream object

```
ofstream oFile;
```

- The file stream object must be connected to a file
  - Connecting a stream to a file is opening the file for the stream

```
iFile.open("myText.txt");
```

Note: in C++98, must be a C-style string, cannot be C++ string! use c_str() to convert C++ string into C string. However, in C++11, could be a C++ string.

Using File Streams

- Use the **input** file stream: use the extraction operator >> and the `getline()` function

```
int bar;
iFile >> bar;
string baz;
getline(iFile, baz);
```

- Use the **output** file stream: use the insertion operator <<

```
oFile << bar;
```

Closing a File

- After using a file, it should be closed

```
file_stream.close();
```

  - This disconnects the stream from the file.
- Why closing a file?
  - Close files to reduce the chance of a file being **corrupted** if the program terminates abnormally.
  - It is important to close an output file if your program later needs to read input from that output file
- The system will automatically close files if you forget as long as your program ends normally
  - … but **explicitly** closing the file is recommended!

Input File Streams

Example

- Consider the following:

```
#include <iostream>
#include <fstream>
using namespace std;
void main() {
  ifstream iFile;
  int bar;
  iFile.open("foo");
  iFile >> bar;
  cout << "The answer is " << bar << ".\n";
  iFile.close();
}
```

- This opens the file named `foo` for reading, and associates it with the input stream object `iFile`.
- Thereafter you can extract input from the file in the same way we did using `cin`.
- If the file named "`foo`" contains the characters "`42`", this program will output:
  `The answer is 42.`

Failed File Streams

- The file stream enters the failed state if:
  - It cannot be opened.
  - You attempt to read past the end of the file.
- A stream's state may be checked by evaluating the stream object:

```
    if(iFile) { ... }
```

  - A stream in the failed state will return false.
- Example

```
iFile.open("a.txt");
if(!iFile) {
    cerr << "Cannot open a.txt\n";
    return -1;
}
```

Example of Reading File

```
while(iFile) {
    getline(iFile, line);
    cout << line << endl;
}
```

Why not good?

How to correct this?

- Normally, after getline reads an entire line, iFile points at the beginning of the next line
- After getline reads the last line, iFile points right before the end of file (think it as a special character)
  - **Note**: iFile is still good! So, the program will issue another getline, which reads nothing
  - So, the program will print an empty line
  - This time, iFile **passes** the end of file and loop terminates

Example of Reading File: Correction

```
while(iFile) {
  getline(iFile, line);
  if(iFile) {
    cout << line << endl;
  }
}
```

Example of Reading File

- Another much simpler (and correct) way

```
while(getline(iFile, line)) {
   cout << line << endl;
}
```

- `istream& getline(istream& is, string& str);`
- Return value: a reference to its parameter `is` (with the value after issuing the current getline).

- <u>Question</u>: why it works?

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String Stream

Motivation

- Suppose that you use the `getline()` function to read an entire line from a file and the result is stored in a string.

```
string line;
getline(iFile, line);
```

- Suppose that the line contains an int followed by a double. We want to read these two numbers from the string `line`.

- We can use input string stream!
  - It reads characters in a string and convert them into values of proper types

String Stream

Motivation

- Suppose we have a string of a book name and an int of its published year. We want to create a string whose first part is the book name and the second part is its published year.
  - Notice that we need to convert the int to a string!

- We can use output string stream!
  - It writes to a string
  - It knows how to convert standard data types into characters and insert them into the string

String Stream

- There are two types of string stream: **input** string stream and **output** string stream.

- C++ defines string stream in the sstream library

```
#include <sstream>
```

- Declare an input string stream object

```
istringstream iStream;
```

- Declare an output string stream object

```
ostringstream oStream;
```

Input String Stream

- When we use input string stream, it is usually assigned a string it will read from.

```
iStream.str(a_string);
```

- We can use extraction operator >> on an input string stream to retrieve the data.

```
istringstream iStream;
int foo;
double bar;
iStream.str(line);
iStream >> foo >> bar;
```

If `line` is the string "`42 3.14`", then

```
foo = 42;
bar = 3.14;
```

Output String Stream

- We can use output string stream to format a string.
  - For example, we might have a collection of numeric values but want their string representation.

- We use insertion operator << to insert characters into an output string stream.
- We fetch the string value of the string stream using the member function `str(void)` of a string stream.

Output String Stream

Example

```
int foo = 512;
int bar = 1024;
string result;
ostringstream oStream;
oStream << foo << " " << bar;
result = oStream.str();
```

result is a string “512 1024”.

Exercise

- Calculate the total expense for fruits.

- Given a data file formatted as follows:

apple 42

banana 25

carrot 33

Write a program that reads all data from the file and sum up the total expense for fruits.

References

- **C++ Primer (4th Edision)**, by *Stanley B. Lippman, Josée Lajoie, Barbara E. Moo*, Addison-Wesley Publishing (2005)
  - Chapter 8.4 File Streams
  - Chapter 8.5 String Streams